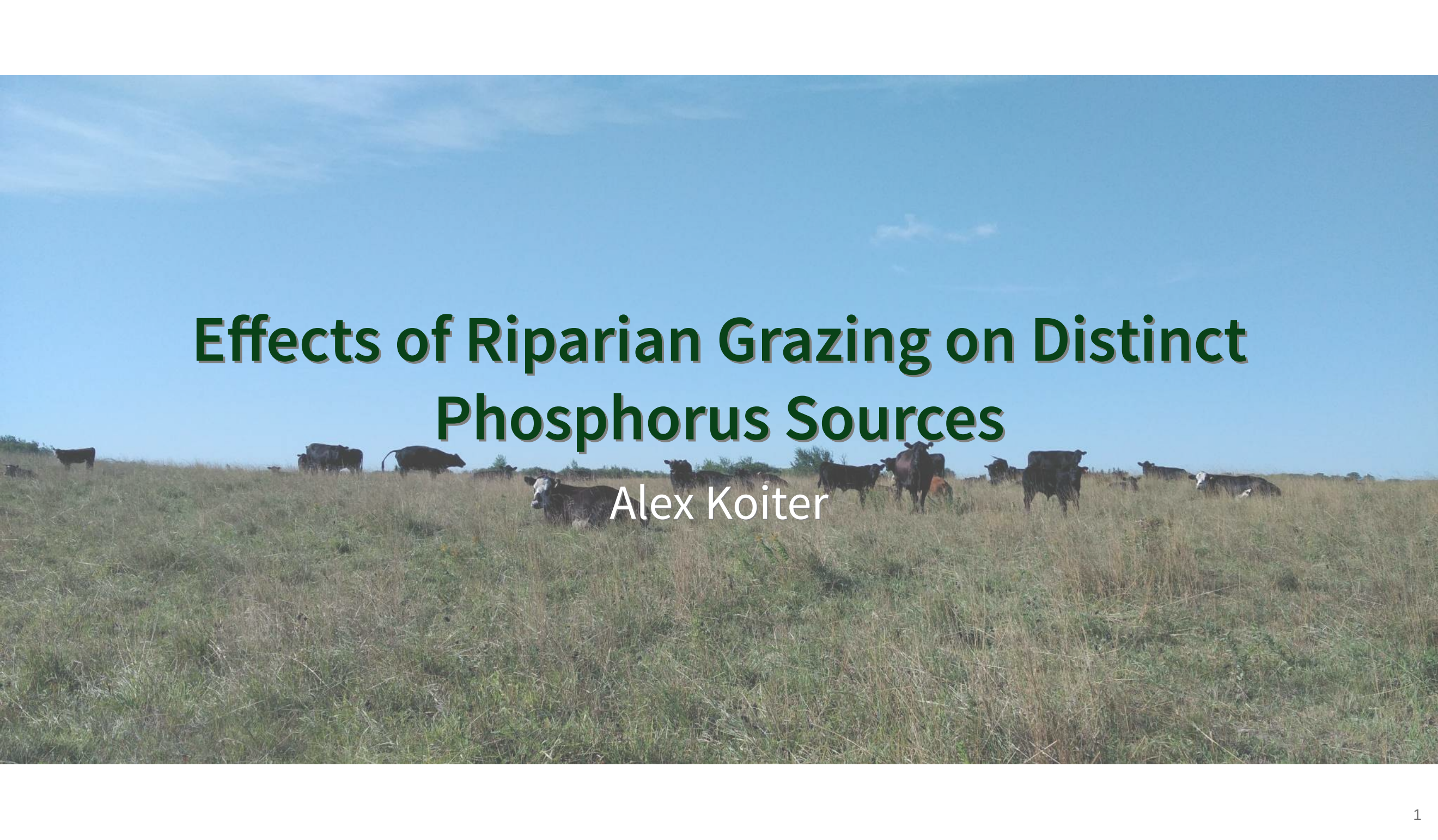


A photograph of a herd of cattle grazing in a field of tall, dry grass under a clear blue sky. The text is overlaid on the image.

Effects of Riparian Grazing on Distinct Phosphorus Sources

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Coauthors

Tamaragh Malone

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**BRANDON
UNIVERSITY**



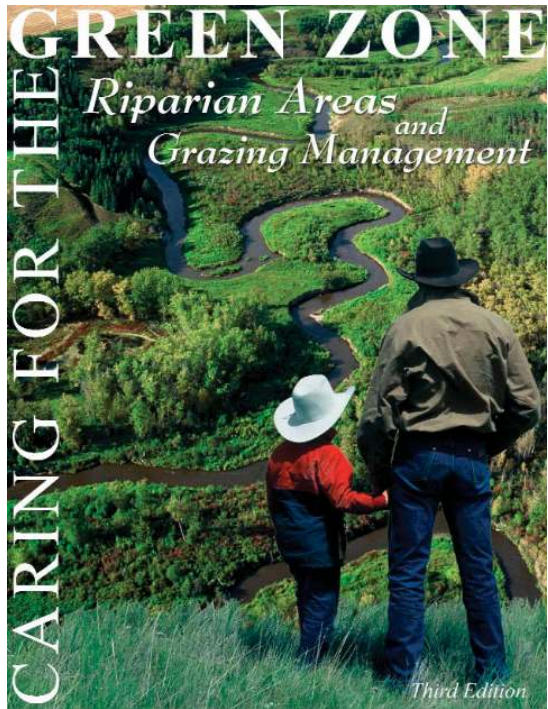
Riparian areas

- Important ecological areas
 - Nesting habitat
 - Transportation corridors
 - Refuge for beneficial plants and insects
- Hydrologically important
 - Bank stabilization
 - Water temperature regulation
 - Reduce flooding risk

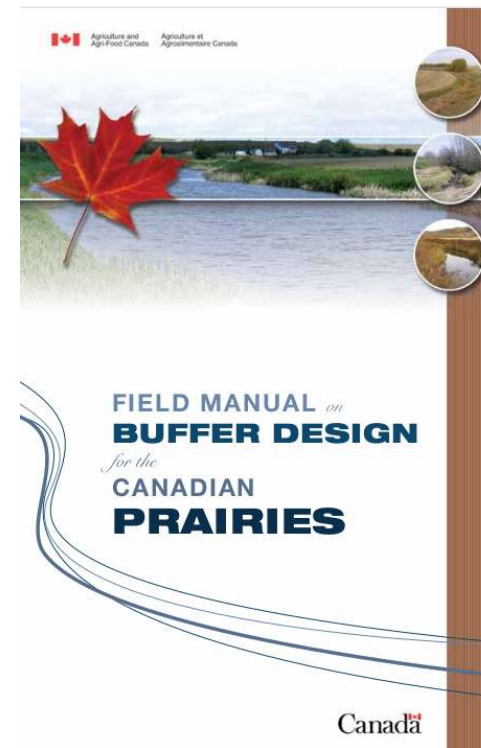


Riparian areas

- Part of the farm
 - Account for 5 to 10% of the farm area
 - Important source of forage
- Riparian areas are maintained to intercept and store nutrients
 - BMP to improve water quality



Cows and Fish Program



Stewart et al. 2010

Riparian areas as filters

Typical prairie riparian area during the snowmelt period

- Most P loading occurs during the spring snowmelt
- Frozen soils and dead and dormant vegetation
 - Filtering capacity is not not very good during the most critical period
- Vegetation can be a source of P



Research objectives

Can we use riparian cattle grazing to reduce the risk of P loss?

1. Assess the distribution of P in two directions

Vertical

- Biomass, Litter, Organic layer, Ah (0-10)

Longitudinal

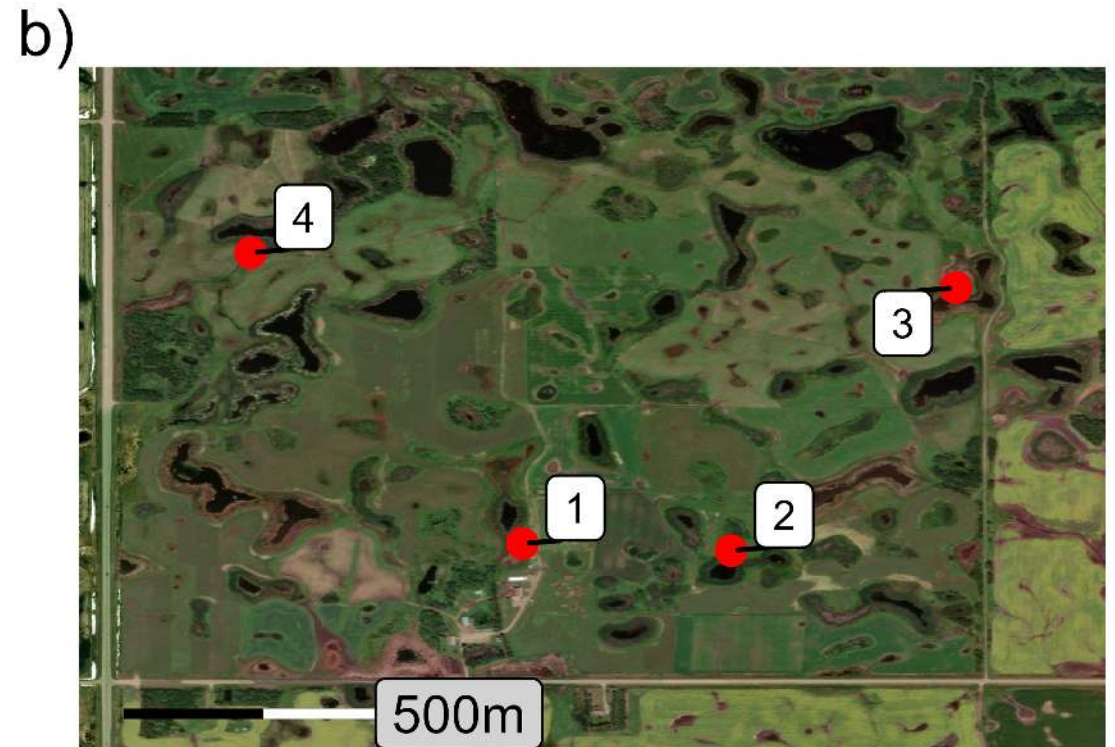
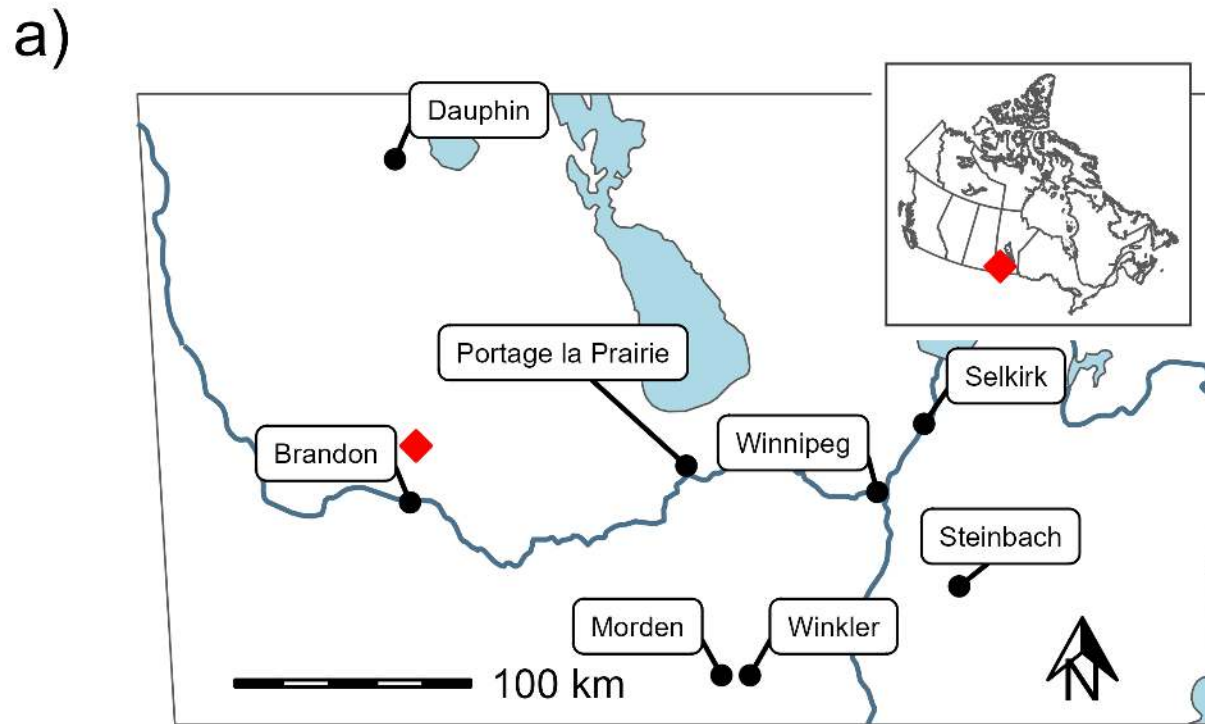
- Upper, mid, lower
2. Quantify the net change in P following grazing



Location

Four riparian areas at the MBFI research farm

- South-western Manitoba



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Treatments

1. Control
 2. Grazing
 3. High-density grazing
 4. Mowing
- Fall grazing
 - Drier soils
 - Protect habitat
 - Extending grazing season
 - Remove P prior to winter



Sampling

Sample collection

- Pre- and post-grazing
- Biomass and litter
 - 0.25 m² quadrat
- Organic and Ah
 - Soil probe



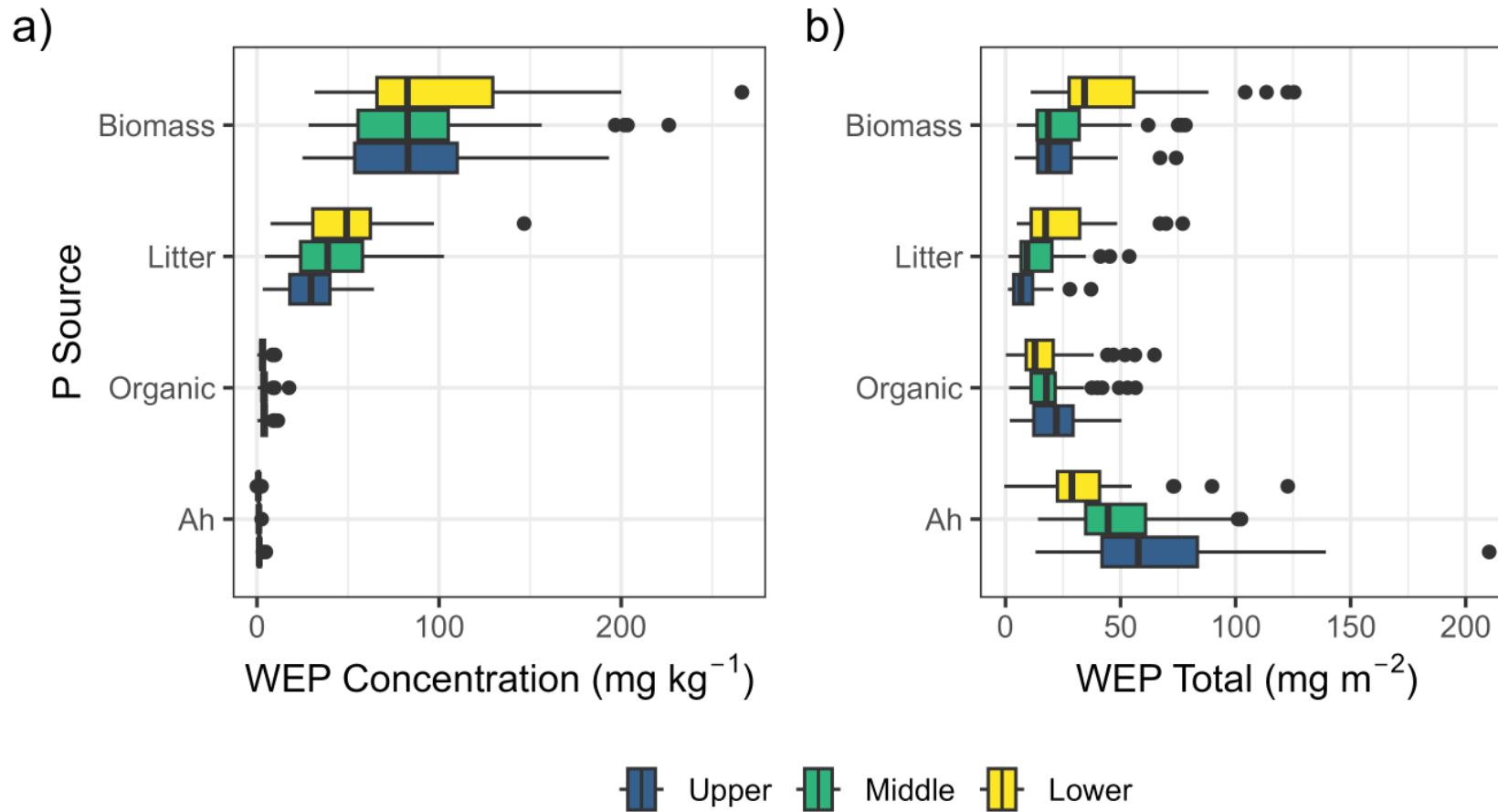
Sample analysis

- Wieghed
- Dried and homogenized
- Water extraction
- Molybdate–ascorbic acid method



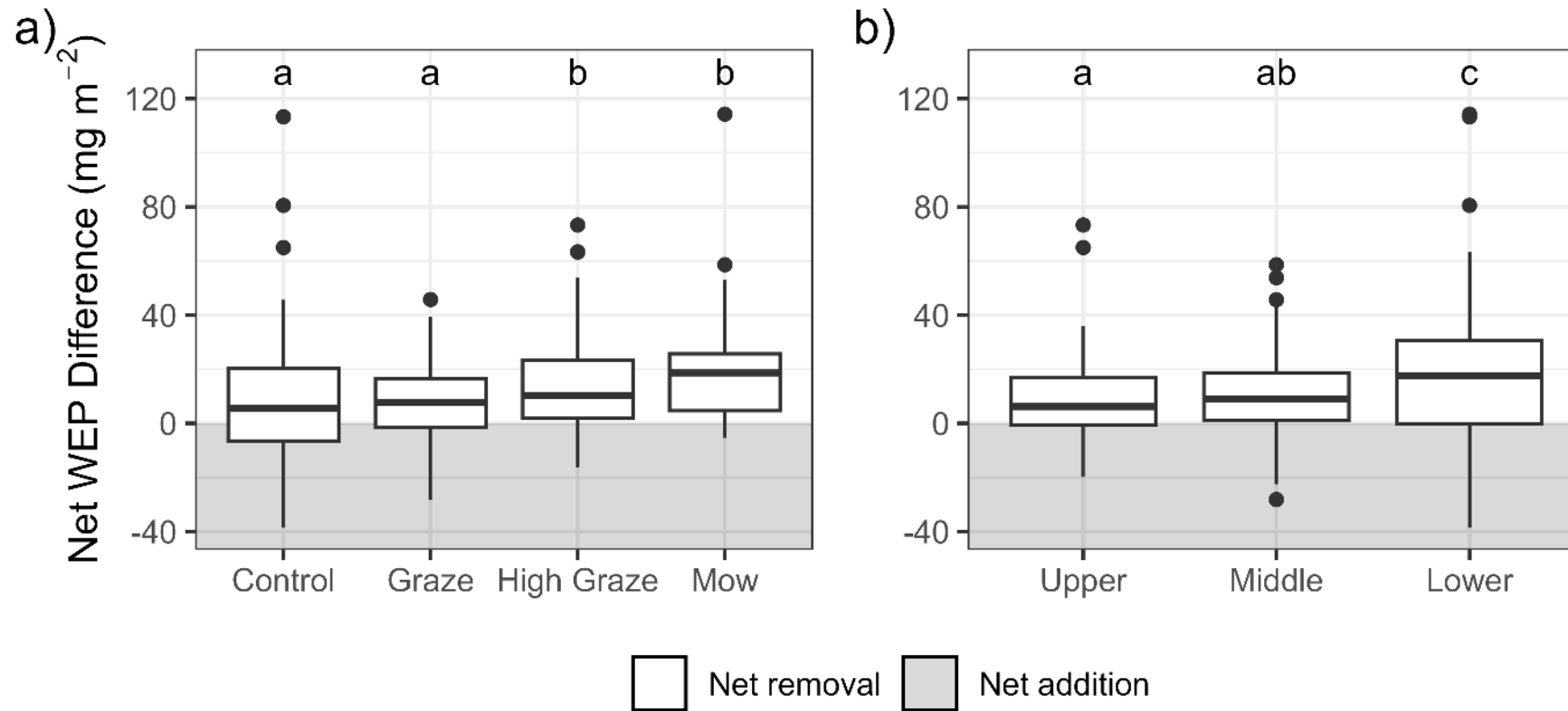
Distribution of P

- Biomass and litter are large sources of P



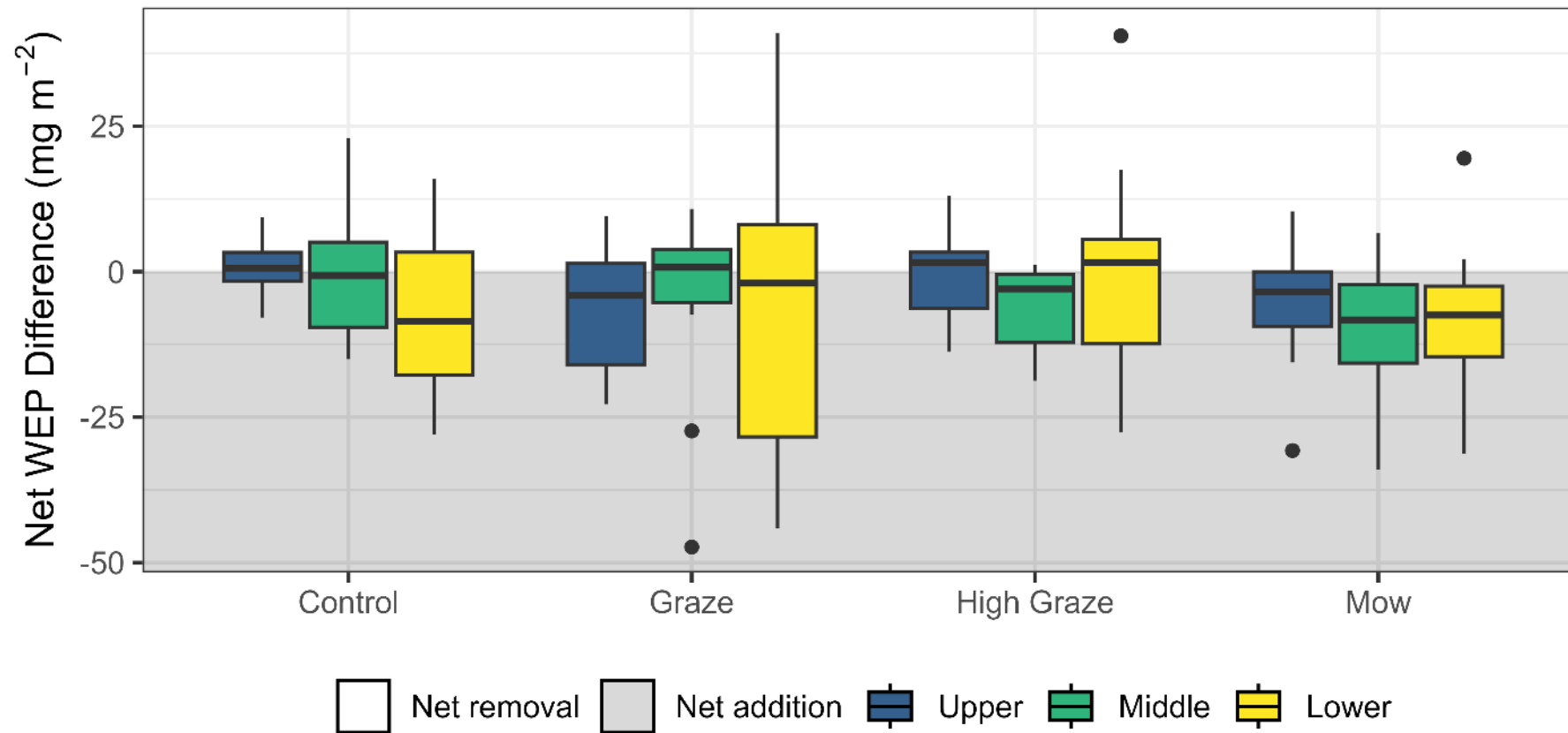
Biomass

- High density grazing and mowing reduced the amount of P
- More P removed in lower slope locations



Litter

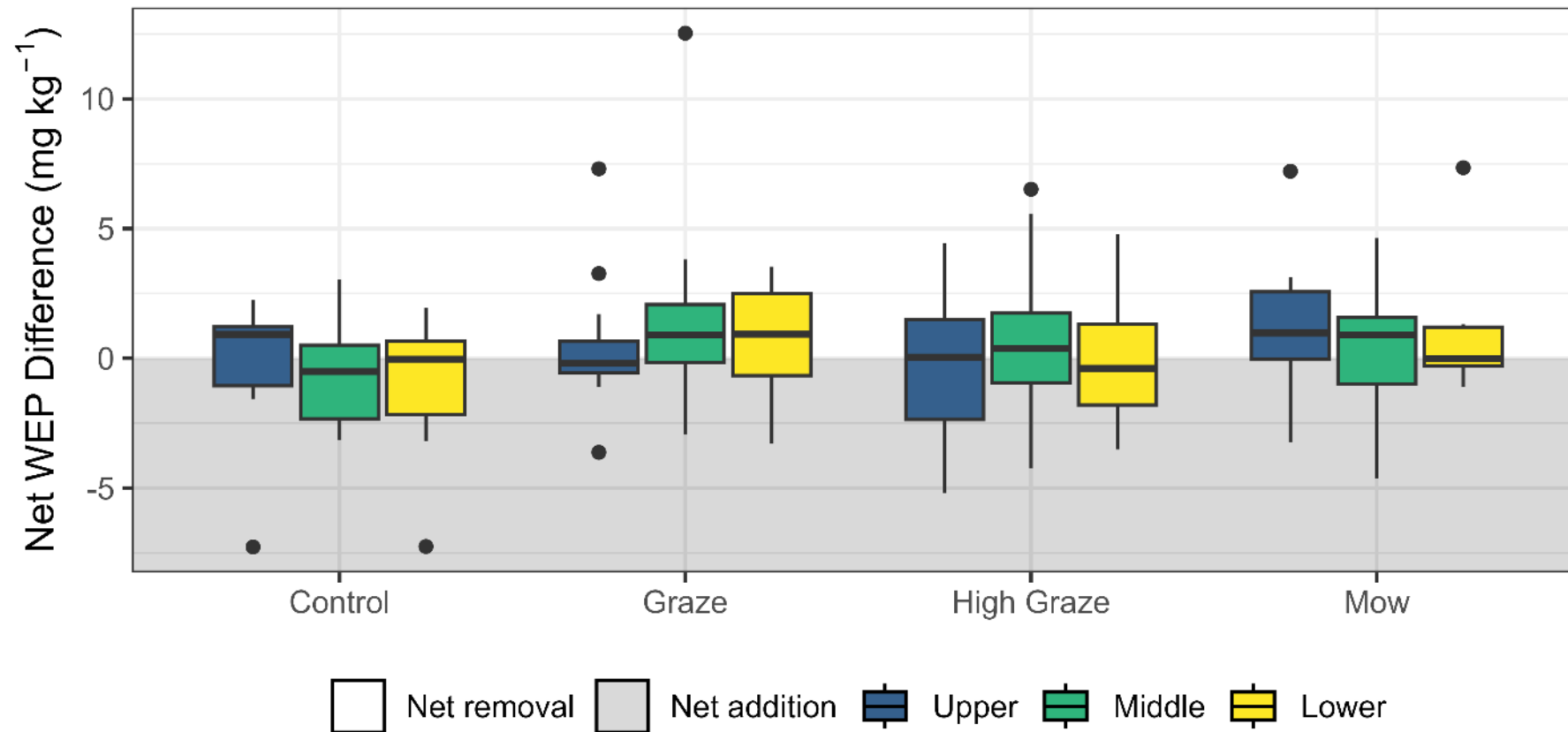
- No treatment effect
- High variability



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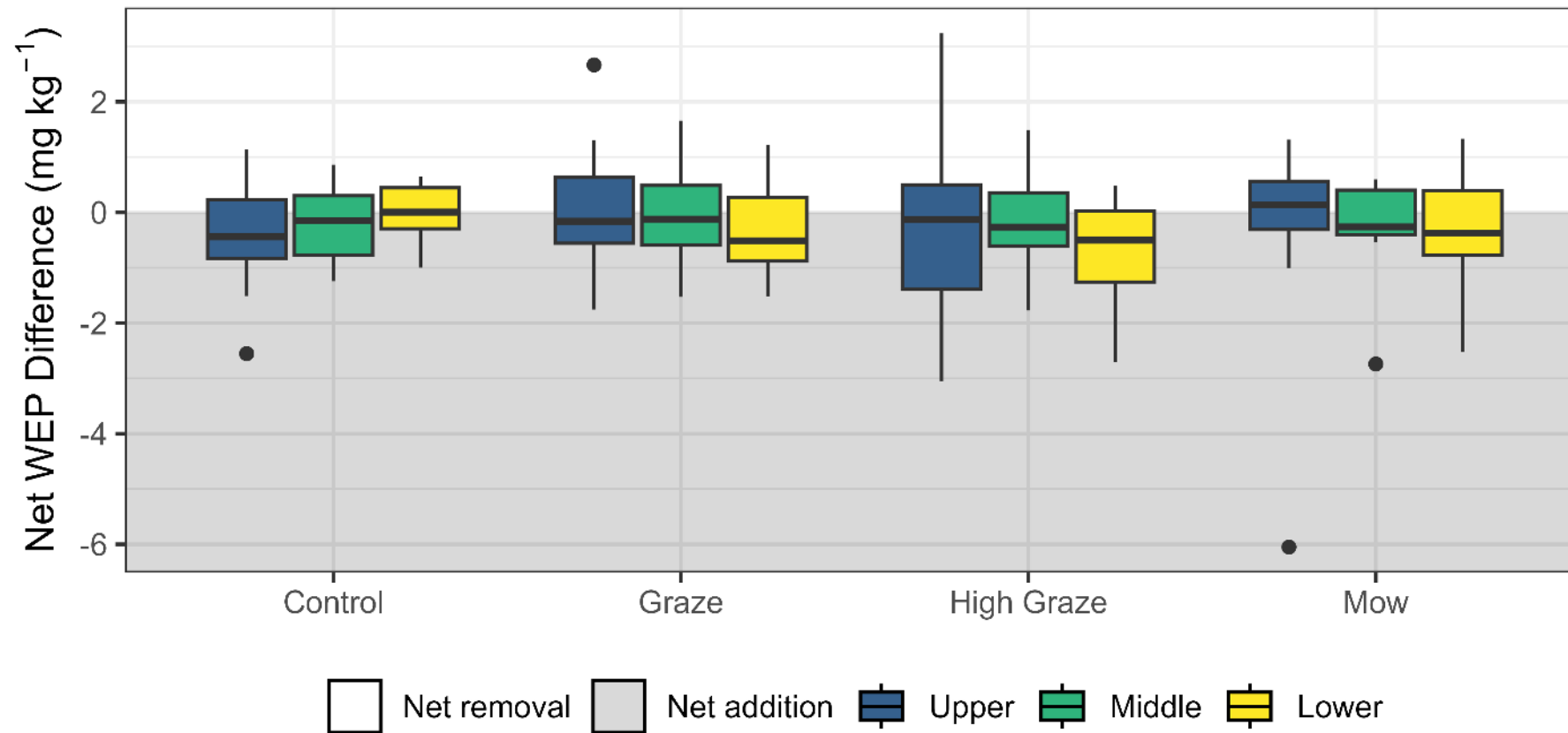
Organic layer

- No treatment effect
- High variability



Ah

- No treatment effect
- High variability



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Variability

- Variability is high
 - Above and below ground
 - Even in the control plots
 - Makes detecting small changes difficult



- Hotspots
 - Manure/urine
 - Plant species
 - Biophysical processes



Conclusions

- Fall riparian grazing reduced the amount of biomass P
 - Reduced risk of P loss during snowmelt
- Variability is high
 - Sampling design needs to accomodate this



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Want to learn more?

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